

Lecture - 8

CO4: Compare various statistical methods of study data samples

CO5: Analyze and evaluation of different sets of data using hypothesis testing.

SPEARMAN'S RANK COEFFICIENT OF CORRELATION

When quantification of variables becomes difficult such beauty of female, leadership ability, knowledge of person etc, then this method of rank correlation is useful which was developed by British psychologist Charles Edward Spearman in 1904. In this method ranks are allotted to each element either in ascending or descending order. The correlation coefficient between these allotted two series of ranks is popularly called as “Spearman’s Rank Correlation” and denoted by “R”. To find out correlation under this method, the following formula is used.

$$R = 1 - \frac{6\sum D^2}{N^3 - N}$$

where, D is the difference of the ranks between paired items in two series and N is the number of pairs of ranks.

In case of tie in ranks or equal ranks: In some cases it may be possible that it becomes necessary to assign same rank to two or more elements or individual or

entries. In such situation, it is customary to give each individual or entry an average rank. For example, if two individuals are ranked equal to 5th place, then both of them are allotted with common rank $(5+6)/2 = 5.5$ and if three are ranked in 5th place, then they are given the rank of $(5+6+7)/3 = 6$. It means where two or more individuals are to be ranked equal, the rank assigned for the purpose of calculating coefficient of correlation is the average of the ranks with these individual or items or entries would have got had they differed slightly with each other. Where equal ranks are assigned to some entries, an adjustment factor is to be added to the value of $6\sum D^2$ in the above formula for calculating the rank coefficient correlation. This adjustment factor is to be added for every repetition of rank.

Adjustment factor = $1/12(m_1^3 - m_1)$ where, m = number of items whose rank are common.

For example, if a particular rank repeated two times, then $m = 2$ and if it repeats three times then $m = 3$ and so on. Hence the above formula can be re-written as follows:

$$R = 1 - \frac{6 * \left[\sum D^2 + \frac{1}{12}(m^3 - m) + \frac{1}{12}(m^3 - m) + \frac{1}{12}(m^3 - m) + \dots \right]}{N^3 - N}$$

Example 1: Find out spearman's coefficient of correlation between the two kinds of assessment of graduate students' performance in a college.

Name of students	A	B	C	D	E	F	G	H	I
Internal Exam	51	68	73	46	50	65	47	38	60
External Exam	49	72	74	44	58	66	50	30	35

Solution:

Name	Internal Exam	Ranks (R ₁)	External Exam	Ranks (R ₂)	D=R ₁ -R ₂	D ²
A	51	5	49	6	-1	1
B	68	2	72	2	0	0
C	73	1	74	1	0	0
D	46	8	44	7	1	1
E	50	6	58	4	2	4
F	65	3	66	3	0	0
G	47	7	50	5	2	4
H	36	9	30	9	0	0
I	60	4	35	8	-4	16
$\sum D^2$						26

$$R = 1 - \frac{6\sum D^2}{N^3 - N} = 1 - \frac{6 * 26}{9^3 - 9} = 1 - \frac{156}{729 - 9} = 1 - \frac{156}{720} = 1 - 0.2167$$

$$= 0.7833$$

Interpretation: From the above calculation it is very clear that there is high degree of positive correlation i.e., **R = 0.7833**, between two exams. It means there is a high degree of positive correlation between the internal exam and external exam of the students.

Example 2: The coefficient of rank correlation of the marks obtained by 10 students in statistics and accountancy was found to be 0.8. It was later discovered that the difference in ranks in the two subjects obtained by one of the students was wrongly taken as 7 instead of 9. Find the correct coefficient of rank correlation.

Solution:

$$R = 1 - \frac{6\sum D^2}{N^3 - N} \Rightarrow 0.8 = 1 - \frac{6\sum D^2}{10^3 - 10} \Rightarrow 0.8 = 1 - \frac{6\sum D^2}{990} \Rightarrow 1 - 0.8 = \frac{6\sum D^2}{990}$$

$$\Rightarrow 0.2 = \frac{6\sum D^2}{990} \Rightarrow 0.2 * 990 = 6\sum D^2 \Rightarrow \sum D^2 = \frac{198}{6} = 33$$

But this is not correct $\sum D^2$ therefore we need to compute correct value Correct $\sum D^2 = 33 - 72 + 92 = 65$

Hence, correct value of rank coefficient of correlation is:

$$R = 1 - \frac{6\sum D^2}{N^3 - N} = 1 - \frac{6 * 65}{990} = 1 - \frac{390}{990} = 1 - 0.394 = 0.606$$

Properties of Coefficient of Correlation:

1. The coefficient of correlation always lies between -1 to $+1$, symbolically it can written as $-1 \leq r \leq 1$.
2. The coefficient of correlation is independent of change of origin and scale.

3. The coefficient of correlation is a pure number and is independent of the units of measurement. It means if X represent say height in inches and Y represent say weights in kgs, then the correlation coefficient will be neither in inches nor in kgs but only a pure number.
4. The coefficient of correlation is the geometric mean of two regression coefficient, symbolically $r = \sqrt{b_{xy} * b_{yx}}$
5. If X and Y are independent variables then coefficient of correlation is zero.